

Statistical Certification of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Dataset Shift

Hassan Waqas Saleem, Rahat Zeeshan Saleem, and Hina Maryam

Abstract—A fallback trigger decides when an automated vehicle should abandon its task for a minimum-risk maneuver. It is usually a hand-tuned threshold without a guarantee, and statistical certification of alert thresholds rarely prices what the fallback then does. We ask how reliable a certified trigger is when the fallback is executed, and what happens under dataset shift. Because a rollout depends only on the tick at which the fallback fires, one reference run plus one forced rollout per decision tick evaluates every trigger, threshold, online method and latency by lookup, checked against direct execution. On replayed Argoverse 2, nuScenes and Waymo logs with reactive traffic and harm defined as an at-fault collision or time-to-collision below 0.95s, we compare a Learn-then-Test-certified geometric trigger with seven baselines. At a 5% target, the certified trigger misses 3.7% of scenarios and exceeds the target in 9.5% of 200 resplits, against 43.5% for the same score tuned without a certificate, which stops less often. Certification buys reliability, not fewer stops. It controls the alert, not the outcome: harm in the intervened run is 6.5%, and 24% of harmful scenarios go unalerted. Confidence and ensemble triggers need 68–71% unnecessary stops. The certificate fails on nuScenes (13.2% missed, wide uncertainty) and in two of six held-out cities, but holds on Waymo (2.7%), so failure is dataset-dependent. One decision tick (0.5s) of latency already breaks the target, and calibrating under that latency restores it only approximately.

Index Terms—Autonomous driving, conformal prediction, minimum-risk maneuver, distribution shift.

I. INTRODUCTION

AUTONOMOUS vehicles that lack confidence in their own situational understanding are expected to fall back on a minimum-risk maneuver (MRM): slow to a stop, pull to the shoulder, or hand control back to a human. When to invoke it is, in most deployed and published systems, a hand-tuned threshold on a scalar confidence or uncertainty score from a perception or prediction module. Such a threshold carries no guarantee, and learned uncertainty estimates are known to degrade under the shift a vehicle meets in a new city or weather. The failure is silent: the system keeps reporting a confidence number that is no longer right.

Manuscript received XX XXXX 2026; revised XX XXXX 2026. This work received no external funding.

H. W. Saleem (corresponding author) is with the School of Electrical Engineering and Computer Science (SEECs), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: hsaleem.msee20seecs@seecs.edu.pk; ORCID: 0009-0003-4461-8005).

R. Z. Saleem is with the NUST Business School (NBS), National University of Sciences and Technology (NUST), H-12, Islamabad 44000, Pakistan (e-mail: rahat.mscm21nbs@student.nust.edu.pk).

H. Maryam is with the State Bank of Pakistan, I.I. Chundrigar Road, Karachi 74000, Pakistan (e-mail: hina.maryam@sbp.org.pk).

Conformal prediction offers a principled alternative, a distribution-free, finite-sample guarantee under exchangeability, and has been applied widely to trajectory forecasting and planning in driving. A guarantee on a prediction, or on a warning, is nevertheless not a guarantee on the consequences of the action it triggers. Whether an alert leads to a fallback is rarely evaluated in closed loop, and triggering is not free: a vehicle that stops in a live lane because its uncertainty grew is not obviously safer than one that continues, and a trigger that fires whenever asked trivially achieves a good miss rate while being useless. Reporting only coverage, or only collision rate, hides this trade-off.

We ask a narrower question than “which trigger is best”: *how reliable is a statistically certified fallback trigger when it is judged through the interventions it actually causes, and what happens to it when the deployment data differ from the calibration data?* Certifying a threshold on a risk score against missed events is not new. Luo *et al.* [27] do so with conformal prediction, and Conformal Decision Theory [31] calibrates decisions through a long-run *average* guarantee. We build on these and do not claim the certification step. An MRM trigger nevertheless differs in kind: it is an irreversible, one-shot decision, its errors are paid in two currencies (missed interventions, and the harm of stopping itself), and a deployed vehicle needs its guarantee before the first mile. To our knowledge, no prior work certifies a fallback trigger and then measures, in closed loop, what the executed fallback does.

We organize the study around four research questions, each tied to a hypothesis fixed before data collection (Section IV-G). **RQ1 (H2)**: does a certificate keep its promise on the closed-loop outcome, and what does it cost in unnecessary stops and induced collisions? **RQ2 (H1)**: do open-loop metrics choose the trigger that closed-loop safety does? **RQ3 (H3, H3-W)**: what happens to a certificate under cross-dataset shift, and how much can unlabeled or a few labeled target scenarios repair? **RQ4**: how sensitive is it to the timing of the intervention?

We make four contributions.

1) **A lookup evaluation protocol for executed fallbacks.**

Under a determinism assumption whose violation we measure, a rollout’s outcome depends only on *when* the fallback fires (Observation 1, Fig. 1). One reference run plus one rollout forced to fire at each decision tick (about 17 per scenario) evaluates every trigger, threshold, online method and latency on identical physics, where direct execution would need over 7,000 rollouts per scenario here. We check the lookup against direct execution

(Section III-E). The forced rollouts also expose induced collisions, route completion, and harm that persists or arises after the fallback fires.

- 2) **What certification buys and what it costs.** Certification is the only design that keeps its target across resplits at a useful operating point, at a measured cost in stops, and it does not win the safety–conservatism trade-off against a tuned threshold (a pre-registered hypothesis, refuted). The certified quantity is the alert, not the outcome: residual harm is larger than the certified miss rate, part of it created by the intervention, and certifying it directly was not achievable at a useful operating point. Confidence and ensemble triggers, the deployed practice, need several times more unnecessary stops than a geometric score.
- 3) **The certificate under real dataset shift.** An ArgoVerse 2 certificate fails on nuScenes but holds, conservatively, on Waymo, so failure is dataset-dependent. The nuScenes data also expose a floor, harm present at the first decision tick, which no trigger can prevent. A density-ratio-weighted certificate that reports its effective sample size repairs part of the gap.
- 4) **Operating-point findings for practitioners.** One decision tick of actuation latency breaks the target, and calibrating under the latency restores it only approximately. Open-loop AUROC ranks triggers only approximately as closed-loop stops do.

Refuted hypotheses are reported as such, and analyses run after the pre-registered verdicts are labeled exploratory.

II. RELATED WORK

A. Conformal prediction for autonomous driving

Conformal prediction [1], [9] constructs prediction sets with a distribution-free, finite-sample marginal coverage guarantee under exchangeability between calibration and test data, and has been applied extensively to trajectory forecasting and to safe planning under uncertainty [10]. Recent work calibrates trajectory-prediction uncertainty separately for each scenario type [11], feeds realized trajectories back into the prediction region [12], certifies the clearance of a selected trajectory with a CVaR margin [13], and couples online calibration with certified navigation in crowds [14]. Lindemann *et al.* [4] embed conformal regions in a model predictive controller for probabilistic safety, a guarantee that collapses when the deployment distribution shifts, which Rahaman *et al.* [23] address with robust conformal prediction in a synthetic multi-agent corridor. SafePath [24] guarantees that a set of candidate paths from an LLM planner contains a safe one, in open-loop (nuScenes) and closed-loop (synthetic traffic) tests. In these works the guarantee is on the *prediction* or on a candidate set, not on a fallback-triggering decision evaluated on replayed driving logs under reactive traffic. Ours is on the decision: the probability that a minimum-risk maneuver fires before a harmful event, evaluated end to end in closed loop.

B. Online and decision-level conformal methods

Adaptive Conformal Inference (ACI) [32] updates the mis-coverage level online so that long-run coverage holds under

arbitrary shift. Dixit *et al.* [5] use it inside a model predictive controller to size a prediction set from delayed observations, a guarantee on the region, not on a fallback decision. Conformal Decision Theory (CDT) [31] carries the idea from sets to decisions: it updates $\lambda_{t+1} = \lambda_t + \eta(\varepsilon - \ell_t)$ so that the long-run average risk converges to ε at rate $O(1/t)$ without distributional assumptions, demonstrated on robot navigation among pedestrians with a continuous distance-based loss. Both give *asymptotic, average* guarantees for a quantity adjusted as data arrive. We target a one-shot, irreversible trigger whose certificate must hold with finite samples before deployment and whose errors include the harm that triggering causes. We adopt both as baselines (Section IV-E), applied scenario by scenario.

Offline risk control now covers non-monotone losses [15], several risks controlled at once with priorities [16], and planning with control barrier functions [17]. Closest to our accounting of what the fallback does, Joshi *et al.* [18] certify the outcome of switching from a baseline policy to a fallback policy, but per instance and independently, with no notion of timing. An MRM trigger is sequential and absorbing: *when* it fires determines every outcome after it.

C. Task-relevant and closed-loop failure detection

Farid *et al.* [25] propose a *task-relevant* monitor that flags a misprediction only when the resulting planning cost is anomalous, with false-positive and false-negative bounds, evaluated open-loop and semi-reactively without an actuated fallback. Their insight is reflected in our labeling: a collision suffered while stopped or struck from behind is an *induced-collision* cost of the fallback, never a missed intervention.

Closest to our certification step, Luo *et al.* [27] tune the threshold of a warning system with Mondrian conformal prediction so that, among situations that are unsafe in the true future *in the absence of corrective action*, fewer than ϵ raise no alert. Their unsafe label is a threshold on a safety score of the logged future, their predictor is Trajectron++, and their alerts are scored against logged data, so the alert triggers nothing. We adopt the same principle and do not claim it as new. The differences that matter are the guarantee (class-conditional there, marginal over scenarios here), the consequence of an alert (none there, an executed maneuver here), and dataset shift and latency, which they do not study (Supplementary Table S1). We implement their certificate as a baseline (Supplementary Section S1).

Sinha *et al.* [38] close the loop differently: a conformal run-time monitor detects when perception leaves the tolerance of a robust MPC controller that keeps a perception-free fallback plan feasible, demonstrated on aircraft taxiing. Their fallback is part of the controller design, whereas ours is an irreversible maneuver whose trigger is the object of study. Atoui and Resende [19] design and validate an MRM controller for sensor failure on highways, taking the decision to engage it as given. Scenario-based simulation is a standard route to evaluating automated vehicle safety [39], [40] and needs a statistical foundation for stopping rules and risk estimates [20], and risk-based filtering can select informative scenarios from

large logs [21]. We replay real logs with reactive traffic and report every rate with a confidence interval.

Fallback triggering also sits within established safety frameworks. Runtime monitors that switch to a trusted fallback, as in [19], [38], follow the runtime-assurance idea of a monitored learned component with a trusted backup. Responsibility-Sensitive Safety [41] gives model-based rules for when a vehicle is at fault, whereas our trigger is a learned score whose threshold is certified from data. SOTIF [42] and UL 4600 [43] frame assurance at the system level, but do not specify how to certify a trigger threshold. Selective prediction [44] studies when a model should abstain at a controlled risk, the same reject-option structure as a trigger, except that here abstention is an executed maneuver with its own cost, and the guarantee is checked under dataset shift. We do not claim priority in any of these areas, and our relation to Luo *et al.* [27] is stated above.

D. Risk-sensitive analysis and closed-loop infrastructure

Chapman *et al.* [28] use Conditional Value-at-Risk to assess the severity, not only the probability, of constraint violation; we control a mean miss rate and leave a tail-severity extension to future work (Section VI). ScenarioNet [29] unifies real driving logs (Argoverse 2, nuScenes, nuPlan, Lyft Level 5) for interactive replay in MetaDrive with logged or reactive (IDM) traffic. Every scenario in Section V is a real logged scenario replayed this way.

III. PROBLEM FORMULATION AND METHOD

A. Fallback triggering as selective prediction

Consider a closed-loop rollout of a scenario X under a fallback policy governed by a scalar threshold λ : at every decision tick the vehicle computes an uncertainty score $u_t(X)$ from a trajectory predictor's output for nearby agents, and engages the minimum-risk maneuver the first time $u_t(X) > \lambda$, remaining engaged for the rest of the episode. Define the closed-loop loss

$$L(X, \lambda) = \mathbb{I}[\text{harm in } X \text{ under } \lambda, \text{ MRM not first}], \quad (1)$$

the *missed-intervention* indicator, and the complementary over-conservatism quantity

$$S(X, \lambda) = \mathbb{I}[\text{MRM fires under } \lambda, X \text{ harm-free at } \lambda = \infty], \quad (2)$$

the *unnecessary stop* indicator, evaluated against the counterfactual rollout of the same scenario with the trigger disabled. The goal is to choose $\hat{\lambda}$ such that

$$\Pr_{X_1, \dots, X_n} \left(\mathbb{E}_X [L(X, \hat{\lambda})] \leq \alpha \right) \geq 1 - \delta, \quad (3)$$

while minimizing $\mathbb{E}_X [S(X, \hat{\lambda})]$, the smallest autonomy loss compatible with the risk target α .

Let $h(X)$ be the first harm step of the reference run with the fallback disabled (∞ if none) and $\tau_\lambda(X)$ the firing step. Because a run under the trigger coincides with the reference run until it fires (Section III-E), $L(X, \lambda) = \mathbb{I}[h(X) \leq \tau_\lambda(X)]$ and $S(X, \lambda) = \mathbb{I}[\tau_\lambda(X) < \infty, h(X) = \infty]$: both alert-level

indicators depend only on the reference run and the firing step. The consequences of the fallback after it fires enter through separate quantities, which we report but do not certify: residual harm (harm anywhere in the executed run), induced collisions, and route completion. Residual harm is not monotone in λ , since firing earlier can itself create harm, and it includes harm after the alert, so (3) does not bound it. Certifying it directly was not achievable at a useful operating point (Supplementary Section S5). Since the running maximum of the score is non-decreasing, τ_λ is non-decreasing in λ and $L(X, \lambda)$ is non-decreasing in λ for every scenario. Equation (3) is therefore a monotone-risk statement, to which Conformal Risk Control [3] applies directly, but with an expectation guarantee. It bounds the calibration-averaged risk, not the probability that a given calibrated threshold exceeds α . We use Learn-then-Test [2] for the high-probability form. For a finite grid $\lambda_1 > \lambda_2 > \dots > \lambda_J$ ordered from least to most conservative, we compute the empirical risk $\hat{R}(\lambda_j) = \frac{1}{n} \sum_i L(X_i, \lambda_j)$ on n calibration scenarios and a Hoeffding–Bentkus p -value for the null hypothesis $H_j : \mathbb{E}[L(\cdot, \lambda_j)] > \alpha$, then apply fixed-sequence testing starting from the most conservative λ . Every λ_j whose null is rejected satisfies (3) simultaneously, with family-wise error controlled at δ , and $\hat{\lambda}$ is the least conservative such λ_j (largest autonomy). The guarantee is therefore about the miss rate of the alert, over scenarios exchangeable with the calibration scenarios, with confidence $1 - \delta$ over the calibration draw. We report the violation frequency of both procedures over resplits rather than assuming it.

The marginal target is weaker than it may sound. With harm in a fraction π of scenarios, a marginal miss probability α leaves a fraction α/π of harmful scenarios unalerted, which is 24% at $\alpha = 0.05$ and $\pi = 0.158$ in our data (Section V-E).

B. Trigger score

At each decision tick the trigger score u_t measures how far a predicted future position of any nearby agent intrudes into the ego vehicle's own near-term extent. Write the ego's constant-heading, constant-speed extrapolation over a horizon of T steps as $\{p_\tau^{\text{ego}}\}_{\tau=1}^T$. This is the same route-agnostic motion model used by the harm metric in Section IV-B, deliberately *not* the vehicle's planned, route-following path, so the trigger cannot credit itself for steering behavior the safety label does not assume. Write a predictor's K -mode forecast for agent i as $\{\hat{p}_\tau^{(i,k)}\}$. Each predicted offset is decomposed into a longitudinal component $a_\tau^{(i,k)}$ (along the ego heading) and a lateral component $\ell_\tau^{(i,k)}$ (perpendicular to it), each compared against a separate margin: a length-based longitudinal margin $m_i^{\text{lon}} = \frac{1}{2}(L_{\text{ego}} + L_i) + c_{\text{lon}}$ and a width-based lateral margin $m_i^{\text{lat}} = \frac{1}{2}(W_{\text{ego}} + W_i) + c_{\text{lat}}$. The intrusion at (i, k, τ) is

$$\rho_\tau^{(i,k)} = \max \left(0, 1 - \sqrt{\left(\frac{a_\tau^{(i,k)}}{m_i^{\text{lon}}} \right)^2 + \left(\frac{\ell_\tau^{(i,k)}}{m_i^{\text{lat}}} \right)^2} \right) m_i^{\text{lat}}, \quad (4)$$

and $u_t = \max_{i,k,\tau} \rho_\tau^{(i,k)}$. A single isotropic clearance radius cannot represent both margins at once: a radius set by width alone risks missing a straight-ahead lead vehicle, and one set

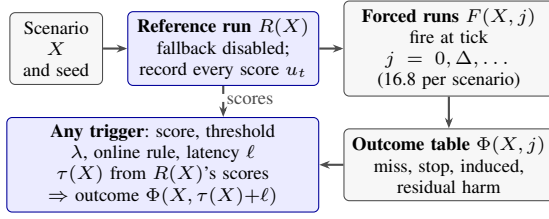


Fig. 1. Lookup evaluation (Observation 1). One reference run and one forced rollout per decision tick are simulated once per scenario. Every trigger, threshold, online update rule and latency is then evaluated by lookup on identical physics, instead of one rollout per combination (over 7,000 per scenario here).

by the half-diagonal risks triggering on adjacent-lane traffic. The decomposition in (4) is therefore a modeling choice made during development on this reasoning. Our ablation (Section V-C) finds that it gives no measurable benefit over a width-only isotropic radius, so we report it as a design option rather than a requirement.

C. Fallback maneuver

On triggering, the vehicle executes the minimum-risk maneuver defined by UN Regulation No. 157 (Automated Lane Keeping Systems) § 5.5.1 [37]: decelerate inside the current lane with a deceleration demand not exceeding 4.0 m/s^2 , lane-keeping control otherwise unchanged, holding the vehicle stationary once stopped. This is enforced by a proportional controller tracking the target speed profile $v(t) = \max(v_0 - 4.0t, 0)$ from the moment of triggering.

D. Certification under covariate shift

The certificate in (3) assumes calibration and deployment scenarios are exchangeable. When the deployment distribution differs by covariate shift, we reweight calibration scenarios by an estimated density ratio $w(x) \propto p_{\text{target}}(x)/p_{\text{source}}(x)$ computed from pre-deployment scenario covariates only (agent counts and speeds, initial gap, route length). A probabilistic domain classifier is trained on unlabeled target scenarios, following weighted conformal prediction [33]. The weighted empirical risk $\hat{R}_w(\lambda) = \sum_i w_i L(X_i, \lambda) / \sum_i w_i$ is tested with the Hoeffding–Bentkus p -value at the Kish effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. As the weights concentrate, n_{eff} shrinks and the test certifies a more conservative threshold, and eventually none at all. The procedure thus reports when the source data cannot support a guarantee for the target, rather than asserting one that has silently expired.

E. Evaluating every trigger by lookup

Fix a scenario X and its random seed. The *reference run* $R(X)$ disables the fallback. For a decision tick j , the *forced run* $F(X, j)$ is identical to $R(X)$ until tick j and then executes the MRM for the rest of the episode. Let $\Phi(X, j)$ denote any outcome of $F(X, j)$ (miss, stop, induced collision, residual harm, route completion) and $\Phi(X, \infty)$ the outcome of $R(X)$. Fig. 1 summarizes the procedure.

Observation 1 (Firing-tick sufficiency): Assume (A1) the simulator, background traffic and predictor are deterministic

functions of the seed and of the ego’s trajectory, so two runs with the same seed and the same ego control up to a step have identical states up to that step; (A2) the trigger’s decision at tick j depends only on the state history up to tick j ; (A3) once the fallback fires it is absorbing, and the control after firing depends only on the state at firing. Then for every trigger with firing tick $\tau(X) \in \{0, \Delta, 2\Delta, \dots\} \cup \{\infty\}$, the run under the trigger is $F(X, \tau(X))$, and every outcome equals $\Phi(X, \tau(X))$. This holds for any score, threshold, online update rule and actuation latency ℓ (for which the run is $F(X, \tau(X) + \ell)$).

Proof: Until τ the control is that of $R(X)$ by (A3) and the states equal those of $R(X)$ by (A1), so by (A2) the trigger’s decisions, and hence τ , are determined by $R(X)$ alone. At τ the control depends only on the state at τ (A3), as in $F(X, \tau)$, and both runs have equal states afterwards by (A1). ■

Two corollaries follow. First, alert-level indicators need no forced run: L and S are functions of $R(X)$ and τ alone. Second, the consequences of the fallback need n_{ticks} forced rollouts per scenario (16.8 on average, at most 26, on the Argoverse 2 test split), after which any number of triggers is evaluated by lookup, where direct execution needs one rollout per trigger, threshold and latency (over 7,000 per scenario here). The observation is elementary and its value is practical: an exhaustive, paired comparison becomes affordable. Assumption (A1) is an idealization, so we measure its violation. Over the first 33,761 forced rollouts (2,000 scenarios), 98.4 % agreed with the reference run to within 1 cm before firing, the pre-fire time-to-collision trace differed in 1.2 %, and the pre-fire harm status in one rollout (0.003 %). We report every analysis on all scenarios and after dropping the 3.6 % with any flagged drift. We also executed four triggers directly, live and without replay, on 252 random test scenarios (1,008 scenario–trigger pairs, one trigger with a one-tick latency). Firing step, miss, unnecessary stop and induced collision agreed with the lookup in all 1,008 pairs (upper bound 0.4 %), and harm anywhere in the run in all but one; route completion differed by at most 0.036 (Supplementary Section S7). The lookup is exact only under (A1), but it matched direct execution within this tolerance.

IV. EXPERIMENTAL SETUP

A. Simulation environment

All rollouts are executed headless in MetaDrive, replaying scenarios converted by ScenarioNet [29] from Argoverse 2 Motion Forecasting [6] and nuScenes [7]. Non-ego traffic is controlled reactively by the Intelligent Driver Model [8], not replayed open-loop from the log. Pure log replay roughly tripled the no-fallback collision rate on a 1,000-scenario diagnostic sample, because logged agents cannot react to an ego vehicle that deviates from its log, precisely what a fallback maneuver produces, and would misrepresent the cost of stopping.

B. Harm labeling

A rollout is labeled harmful using the nuPlan closed-loop safety definition [30]: an at-fault collision, or a time-to-collision below 0.95 s computed over a 3.0 s horizon at

0.1 s resolution, not evaluated while the ego is stationary or against tracks behind it. Fault attribution follows the nuPlan convention. A collision the ego suffers while stationary, or while struck from behind, is not at-fault and never counts as a missed intervention. It is recorded as an *induced collision*, the measurable cost of a false or premature trigger.

C. Predictor

The trajectory predictor is AutoBot [35], one of many learned multi-modal forecasters [22] (6 modes, 2.1 s history, 6 s horizon), trained with the UniTraj framework [36] on the Argoverse 2 Motion Forecasting training split (minADE₆ = 1.09 m on held-out validation scenes). It runs online inside the closed loop, so its input is the actual, possibly ego-deviated, simulator history, with the UniTraj preprocessing unchanged. Vehicles are predicted by the network. Pedestrians and cyclists, which the checkpoint was not trained on, receive constant-velocity forecasts.

D. Datasets and splits

All scenarios come from Argoverse 2 validation, disjoint from the predictor’s training data. They are split deterministically by scenario identifier into development (14,990), calibration (4,971) and test (5,027) sets, and design choices were made on the development split only. The shifted target is nuScenes (9,041 scenarios of 8.1 s, Boston and Singapore, versus Argoverse 2’s six U.S. cities). Scenarios in which the ego moves less than 10 m are excluded, as in ScenarioNet. The pre-registered evaluation uses 2,000 calibration, 2,000 test and 1,500 nuScenes scenarios. Ten Argoverse 2 scenarios (six calibration, four test) whose per-agent records could not be matched after a re-scoring pass are dropped from every analysis, leaving 1,994 calibration, 1,996 test and 1,504 nuScenes scenarios. A second shifted target, 1,000 Waymo Open Motion [26] validation scenarios (9.1 s, 1 s of history), was collected under a separate pre-registered amendment (Section V-F). Decisions are taken at 2 Hz, with identical miss rates at 2 and 10 Hz on a paired development comparison and a 10 Hz replication on 100 test scenarios.

E. Baselines

All methods use the same predictor, the same MRM and the same test scenarios, and are evaluated by lookup (Section III-E).

- **T1 — confidence threshold** (deployed practice): fire when $1 - \max_k p_k$ of the network-predicted vehicle whose predicted modes come closest to the ego exceeds a threshold. Agents the network does not predict (pedestrians, cyclists, distant vehicles) are excluded from this score. The threshold is tuned on calibration data to the most permissive value with empirical miss rate $\leq \alpha$. A variant that uses the nearest agent ahead is also evaluated.
- **T2 — ensemble threshold**: the same tuning, applied to the spread of endpoint predictions across five Monte Carlo dropout passes [34].

- **T3 — open-loop conformal**: inflate every predicted mode by the split-conformal radius of the predictor’s 3 s error on held-out Argoverse 2 logs at level α [1], and fire when the inflated region touches the ego. This is the guarantee-on-the-prediction design that the paper argues is insufficient.
- **Tuned geometric**: our score with an empirically tuned threshold and no certificate. It isolates the contribution of the calibrator.
- **CDT** [31] and **ACI** [32]: online decision- and prediction-level adaptation, updated after every scenario. Each uses the step size that performs best on test, which favors the baselines. Choosing it on calibration data instead gives CDT 4.6 % missed and 25.1 % stops, and ACI the same result.
- **Oracle**: the best threshold in hindsight on test data. It is not deployable and is reported as an upper bound.

F. Metrics and statistics

For every method we report the missed-intervention, unnecessary-stop and fallback-induced-collision rates, autonomy (the fraction of scenarios completed without the fallback) and route completion, with Wilson 95 % intervals. Validity of each certificate is assessed over 200 random half-and-half resplits, without replacement, of the pooled 3,990 calibration and test scenarios. A valid certificate keeps the frequency of test miss rate above α at or below δ . The resplits overlap, so they are not independent draws: the frequency estimates the violation probability for this pool, with Monte Carlo error (Wilson 95 % interval [6.2, 14.4] for 9.5 %, [36.8, 50.4] for 43.5 %). Scenarios from one place may be dependent. Resplitting whole spatial groups (ego starts within 150 m, a proxy) raises LTT’s violation frequency to 17.0 % [12.4, 22.8], above δ , although group bootstrap intervals for the miss rates barely widen (Supplementary Section S12). Harm uses the nuPlan definition above, and Section V-G repeats the key comparison under three alternatives.

G. Pre-registration and reproducibility

The harm definition, $\alpha = 0.05$, $\delta = 0.10$, and the refutation criteria for three hypotheses were recorded in a version-controlled note before any calibration or test data were collected: **H1**, that open-loop metrics rank triggers differently from closed-loop outcomes; **H2**, that a certified trigger achieves a better safety–conservatism trade-off than a tuned fixed threshold; and **H3**, that a threshold certified on Argoverse 2 loses its guarantee on nuScenes. A later amendment, written before any Waymo outcome existed, added **H3-W**, the same hypothesis on Waymo, and a few-label recalibration study. The note carries further dated amendments (a corrected confidence baseline, the exactness measurements, outcome-level certification, robustness checks and a direct-execution check), each written before the corresponding outcomes were evaluated, and it labels the analyses run afterwards as post hoc.

Data and code availability. Code, the pre-registration note with all amendments, configuration files and per-scenario

rollout records are available at <https://github.com/hwaqas160/paper03-risk-fallback> and archived at <https://doi.org/10.5281/zenodo.23054382>. Each scenario's record contains every score trace and outcome, so every table can be regenerated without re-simulation.

V. RESULTS

All results use $\alpha = 0.05$ and $\delta = 0.10$, 1,994 calibration and 1,996 test scenarios from Argoverse 2, and 1,504 nuScenes scenarios, with Wilson 95 % intervals unless stated. Dropping the 3.6 % of scenarios with any flagged simulator drift changes no conclusion below. Sections V-A–V-D are the pre-registered analyses. Of the subsections that follow, the Waymo recalibration (R1–R4) is pre-registered; the consequences of the intervention, the class-conditional comparison, the reverse shift, few-label recalibration and the robustness checks (Section V-G) are post hoc or partly so, and are labeled. Full details of the exploratory analyses are in the Supplementary Material.

A. Does a certificate keep its promise, and what does it cost? (RQ1)

Table I gives every method at its operating point on the Argoverse 2 test set. The certified trigger (LTT) fails to pre-empt harm in 3.7 % of all scenarios [95 % CI 2.9–4.6], below the 5 % target, which is 24 % of the scenarios that are harmful, with 28.5 % unnecessary stops and 2.6 % fallback-induced collisions. The last column repeats the calibration on 200 random resplits and reports how often the realized test miss rate exceeds α . A valid certificate keeps this at or below $\delta = 10$ %. LTT does (9.5 %). The same geometric score with a threshold tuned empirically, and its Conformal Risk Control (CRC) counterpart, which guarantees only an expectation, exceed the target in 43.5 % and 41.0 % of resplits, and the tuned confidence trigger in 50.5 %. Certification is therefore the only design here that keeps its target reliably at a useful operating point. Fig. 2 plots every method against the geometric score's own curve as its threshold varies, and Fig. 3 shows the resplit distributions behind the violation column. CRC and empirical tuning select the same threshold on this calibration set. CRC requires $\hat{R} \leq \alpha - (1 - \alpha)/n = 4.95$ % instead of 5 %, and the calibration risk jumps from 4.81 % to 5.02 % between adjacent grid points, so no threshold falls in between.

The certificate has a price, and it does not win the trade-off. The tuned geometric threshold and CDT both stop less often than LTT (Table I; paired bootstrap difference -2.5 points, 95 % CI $[-3.2, -1.8]$ against tuned). Hypothesis H2 required no baseline to be within 3 points of the certified trigger. It is **refuted**. The supportable claim is narrower: among the designs that keep their promise at a useful operating point, the certified one is the only one that reliably meets it. Open-loop conformal never violates, but fires in 73.8 % of scenarios.

Deployed-practice triggers are far more conservative. Tuned to the same miss target, confidence and MC-dropout stop 40–45 points more often than the certified geometric trigger (Fig. 2; paired CIs exclude zero by a wide margin). Open-loop conformal reaches a miss rate of 0.0 % only by

TABLE I
CLOSED-LOOP OUTCOMES ON THE ARGOVERSE 2 TEST SET AT $\alpha = 0.05$.
RATES IN %, UNNECESSARY-STOP RATE WITH ITS 95 % WILSON CI
WHERE AVAILABLE. LAST COLUMN: FRACTION OF 200
CALIBRATION/TEST RESPLITS IN WHICH THE TEST MISS RATE EXCEEDS α
(VALID IF ≤ 10).

Method	Miss	Unnec.	95 % CI	Induced	Auton.	Viol.
No fallback	15.1	0.0	[0.0, 0.2]	0.0	100	–
Always fire	0.0	84.9	[83.2, 86.4]	6.5	0	–
T1 confidence	4.6	68.3	[66.2, 70.3]	4.0	20.3	50.5
T2 MC-dropout spread	4.0	70.8	[68.8, 72.7]	4.9	17.6	–
T3 open-loop conformal	0.0	73.8	[71.8, 75.7]	5.5	11.1	0.0
Tuned geometric (no cert.)	4.3	26.0	[24.1, 28.0]	2.3	61.4	43.5
CRC (geometric)	4.3	26.0	[24.1, 28.0]	2.3	61.4	41.0
LTT (geometric, certified)	3.7	28.5	[26.5, 30.5]	2.6	58.5	9.5
CDT (online)	4.9	24.1	–	2.2	63.9	–
ACI (online)	1.3	45.0	–	3.5	40.2	–
Oracle (hindsight)	5.0	23.6	–	2.2	64.2	–

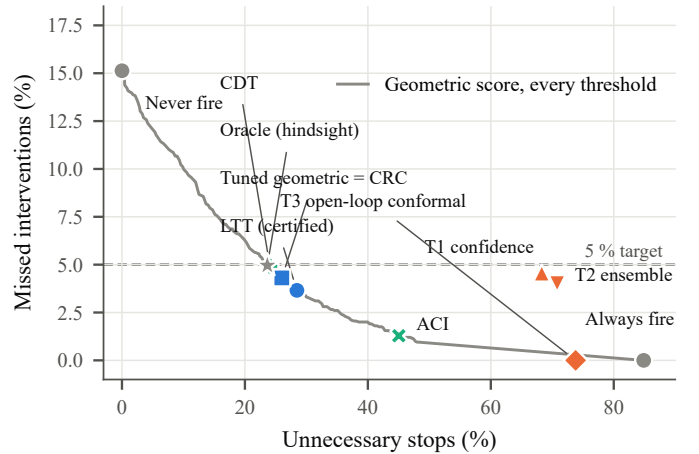


Fig. 2. Every method at its operating point (Table I), against the geometric score's own miss-versus-stops curve as its threshold varies. On this split most methods sit below the dashed $\alpha = 0.05$ line; the geometric methods do so at well under half the stops of the confidence, ensemble and open-loop triggers. Which of them keeps the target across resplits is shown in Fig. 3.

firing in 73.8 % of scenarios: valid in that it never violates, useless in that the vehicle abandons its task three times in four. ACI's 1.3 % misses at 45 % stops fit the same pattern, a guarantee on the prediction not translating into a useful decision.

B. Do open-loop metrics choose the closed-loop trigger? (RQ2)

Table II ranks six trigger scores by open-loop AUROC of their scenario peak for predicting harm, and by closed-loop unnecessary stops at each score's certified threshold. Rankings agree at the top and diverge below it: Kendall's $\tau = 0.60$ on Argoverse 2, 0.47 on nuScenes and 0.73 on Waymo. The top-ranked score is the same on Argoverse 2 and on Waymo, and differs on nuScenes by only 0.2 points of stops. The pre-registered rule refutes H1 only if $\tau \geq 0.8$ and the top score coincides in every setting, so H1 is not refuted, but the evidence is weak. The clearest disagreement is at the bottom: the confidence score has AUROC 0.45, below chance, so higher confidence-based uncertainty is on average associated

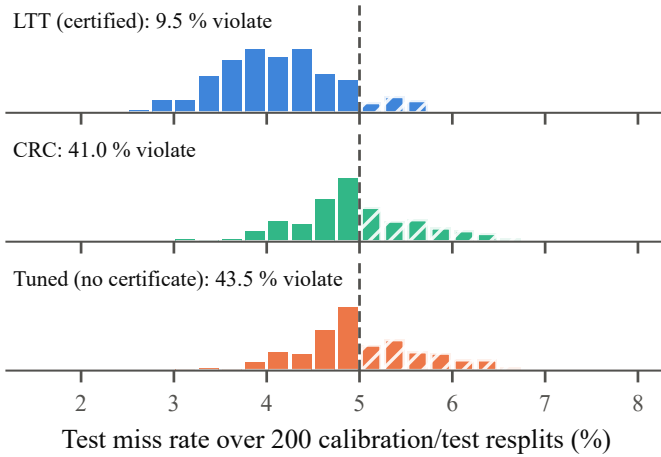


Fig. 3. Test miss rate over 200 calibration/test resplits. Dashed line: the 5 % target; hatched bars: resplits that violate it. A valid certificate keeps violations at or below $\delta = 10$ %. LTT's high-probability guarantee does (9.5 %); CRC's expectation guarantee and empirical tuning do not.

TABLE II
OPEN-LOOP AUROC VERSUS CLOSED-LOOP UNNECESSARY STOPS AT THE CERTIFIED THRESHOLD, ARGOVERSE 2 TEST.

Score	AUROC	Unnec. stops (%)
Geometric, isotropic radius	0.843	27.5
Geometric, length-aware box	0.832	28.5
Predictor gap	0.817	31.8
Geometric, route corridor	0.577	84.9
Confidence	0.448	75.8
Ensemble spread	0.442	72.3

with *less* harm, yet a threshold on it still reaches the miss target, at 75.8 % stops, by firing indiscriminately.

C. Ablations and latency (RQ4)

Table III shows that two components matter and one does not. A route-following corridor makes the trigger fire in essentially every scenario (84.9 %), consistent with the route reaching conflict points that the kinematic harm label does not count, and replacing geometry with confidence or ensemble spread costs 44–47 points of stops. The length-aware box does not matter: a width-only isotropic radius performs the same (27.5 % versus 28.5 %). One decision tick (0.5 s) of delay already lifts the miss rate above the target (5.3 %) and two ticks give 9.1 %, while stops fall (Fig. 4b), and the zero-latency certificate violates the target in 92 % and 100 % of resplits. Calibrating under the deployed latency, which the lookup supports without new simulation, gives 3.6–4.1 % missed at 34.4 % and 42.5 % stops, but its violation frequency, 12.0 % and 11.5 %, exceeds δ as pre-registered, so we do not claim validity there (Supplementary Section S9).

D. Dataset shift (RQ3)

The Argoverse 2 certificate does not transfer. On nuScenes the threshold certified at a 5 % target misses 13.2 % [11.6, 15.0], 2.6 times the target and with an interval that excludes it. The pre-registered hypothesis H3 is **supported** (15.2 % after

TABLE III
ABLATIONS ON THE ARGOVERSE 2 TEST SET, CERTIFIED BY LTT.

Variant	Miss (%)	Unnec. stops (%)
Full score, LTT, no latency	3.7	28.5
Route-following ego corridor	0.0	84.9
Width-only isotropic radius	3.6	27.5
Confidence instead of geometry	3.6	75.8
Ensemble spread instead of geometry	3.6	72.3
CRC instead of LTT	4.3	26.0
Empirical tuning instead of LTT	4.3	26.0
+1 decision tick latency (0.5 s)	5.3	26.3
+2 decision ticks latency (1.0 s)	9.1	23.4

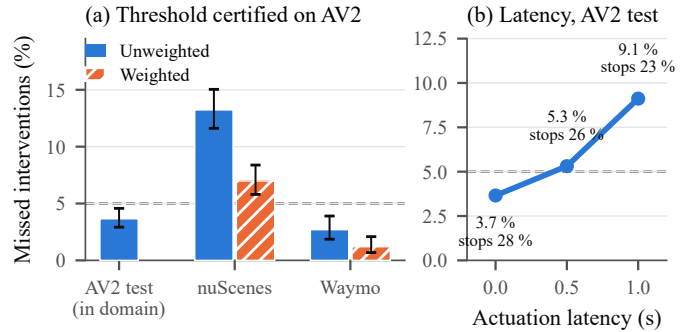


Fig. 4. (a) Miss rate of the Argoverse 2-certified threshold deployed unchanged on nuScenes and Waymo, unweighted and with the density-ratio repair (Section III-D); error bars are 95 % Wilson CIs. (b) Miss rate and unnecessary stops as actuation latency increases (Table III).

dropping drift scenarios). Part of this is a floor rather than shift: 5.0 % of nuScenes scenarios already contain harm at the first decision tick, which even a trigger that always fires cannot prevent (0.1 % on Argoverse 2). That floor equals the target, so no threshold can be certified at $\alpha = 0.05$ on nuScenes. With those scenarios excluded (post hoc), the shifted miss rate is 9.2 % [7.8, 10.8], still above the target. These intervals treat scenarios as independent, but the 1,504 nuScenes scenarios come from only 37 scenes. Resampling scenes widens the interval to [6.1, 21.0], which still excludes the target, while a second, disjoint set of 23 scenes gives 9.5 % with scene-level interval [3.9, 16.5], which does not: the direction of the failure replicates but its size is uncertain (Supplementary Section S6). A domain classifier on the covariates separates Argoverse 2 from nuScenes (cross-validated AUROC 0.97, $n_{\text{eff}} = 66$) far more easily than from Waymo (0.91, 288) and not at all from its own test split (0.50), so the certificate failed under the larger measured shift and held under the smaller, but two targets cannot establish a trend. Six further natural shifts, the cities of Argoverse 2 held out one at a time, add evidence. The certificate met the target in four of six (67 %), below the 80 % we pre-registered, so that claim is refuted. It failed in Miami (6.1 % missed [4.8, 7.8]) and Washington, DC (5.3 %), the two cities with the highest harm rates, and across these cities and the two datasets the held-out miss rate tracks the target's harm rate (Spearman 0.90, $n = 8$, post hoc), which suggests a shift in the base rate of harm as a common cause (Supplementary Section S10).

Density-ratio weighting, using only unlabeled target covari-

TABLE IV

A THRESHOLD CERTIFIED ON ARGOVERSE 2 (LTT, $\alpha = 0.05$) DEPLOYED UNCHANGED ON nuSCENES AND WAYMO. “AVOIDABLE” EXCLUDES THE 5.0 % OF nuSCENES SCENARIOS WITH HARM AT THE FIRST DECISION TICK (POST HOC). WEIGHTED: DENSITY-RATIO REPAIR, NO TARGET LABELS.

Setting	Miss (%) [95 % CI]	Unnec. (%)	n
Argoverse 2 test (in domain)	3.7 [2.9, 4.6]	28.5	1,996
nuScenes, unweighted	13.2 [11.6, 15.0]	26.1	1,504
nuScenes, weighted	7.0 [5.8, 8.4]	52.4	1,504
nuScenes, oracle threshold	5.0	77.9	1,504
Avoidable, unweighted	9.2 [7.8, 10.8]	27.1	1,429
Avoidable, weighted ($n_{\text{eff}} = 68$)	2.2 [1.6, 3.1]	54.5	1,429
Waymo, unweighted	2.7 [1.9, 3.9]	45.1	1,000
Waymo, weighted	1.2 [0.7, 2.1]	50.2	1,000
Waymo, oracle threshold	5.0	33.9	1,000

ates, moves the miss rate from 13.2 % to 7.0 %, still above the target, and doubles unnecessary stops (26.1 % to 52.4 %). On the avoidable subset it reaches 2.2 %, below the target, but its effective sample size is only 68, so the certificate is achieved almost entirely through conservatism (54.5 % stops).

E. What the intervention does, and a class-conditional certificate (post hoc)

The certified quantity is the alert. Supplementary Section S14 partitions every scenario by what happens after the fallback fires. At the certified threshold on Argoverse 2, 3.7 % of scenarios are missed, in a further 1.7 % the fallback fires before harm but harm still occurs, and in 9.8 % harm is prevented. In a further 1.1 % of all scenarios, which are harmless without a fallback, the intervened run contains harm (nuPlan definition), and 2.6 % suffer an induced collision. Harm in the intervened run is therefore 6.5 % (15.1 % without a fallback), almost twice the certified miss rate. On the fresh replication split it is 5.0 % [3.7, 6.6] against 3.4 % missed. A trigger that always fires creates harm in 3.1 % of scenarios. On nuScenes the same threshold leaves 15.6 % residual harm. Induced collisions are other vehicles striking the ego, at low to moderate speed (median speed difference 2.4 m/s), typically long after it has slowed (median 6.5 s after firing) and often where it starts at standstill, so this cost may partly reflect simulated traffic that does not yield to a stationary vehicle (Supplementary Section S6). The fallback removes a large share of harm, but the certificate does not speak to what remains, and certifying residual harm directly at 5 % required firing in nearly every scenario and still failed validity (Supplementary Section S5).

Class-conditional certification and reverse shift. A Luo-style class-conditional certificate at $\epsilon = 0.05$ is equivalent to always firing (84.9 % stops), and at the marginal-matched $\epsilon = 0.32$ its conditional miss rate on nuScenes is 66 % against its 32 % target (Supplementary Section S1). Certifying on nuScenes and deploying on Argoverse 2 transfers safely but conservatively (miss 1.9–2.0 %, Section S2), a single-pair observation.

F. Second shifted dataset and recalibration

Waymo Open Motion. Table IV and Fig. 4a deploy the same Argoverse 2-certified threshold, unchanged, on 1,000 Waymo Open Motion validation scenarios [26] (9.1 s, 1 s of history). It misses 2.7 % [1.9, 3.9], below the 5 % target. Hypothesis H3-W was pre-registered with the expectation that this interval would exclude α from above, as on nuScenes; it does exclude α , but from below, so H3-W is **refuted** and failure under shift is dataset-dependent. A density-ratio-weighted repair, applied for consistency even though nothing here needed repairing, lowers the miss rate further to 1.2 % [0.7, 2.1] at a small increase in stops (45.1 % to 50.2 %).

Waymo recalibration (pre-registered R1–R4). We draw k labeled Waymo scenes at random (200 draws) and evaluate on the rest, a design pre-registered before any Waymo outcome was read (full table and the exploratory nuScenes counterpart in Supplementary Section S3). Uncertified tuning violates the target in 69.5 % and 59 % of draws with 40 and 60 target scenes (**R1 holds**: violation ≥ 50 % at both). Weighting the Argoverse 2 calibration set with 60 Waymo scenes ($n_{\text{eff}} = 232$) keeps validity (0 % violation, **R2 holds**) at a cost of 21.3 points of unnecessary stops over the in-domain rate (**R3 holds**). **R4**, which restates H3-W, is refuted for the same reason: the source certificate already holds on Waymo, so there was nothing to repair. Few labeled target scenes are therefore dangerous without a certificate even when the source certificate transfers.

Sensitivity. Non-reactive replay traffic, a 2.0 m/s² comfort deceleration and 10 Hz decisions leave the certified miss rate below the target and every conclusion above unchanged (Supplementary Section S4). Thresholds certified on reactive traffic and deployed on replay traffic still miss only 3.7 % [2.2, 6.1], with 35.4 % stops against 71.7 % for the tuned confidence trigger (Supplementary Section S11).

G. Robustness checks (post hoc)

Two objections are addressed on data not used above, in checks pre-registered as post hoc before running.

Does the geometric score’s advantage depend on matching the harm label? The score and the nuPlan label are both proximity- and time-to-collision-based, so a reviewer could suspect the comparison with confidence and ensemble scores is unfair by construction. Table V re-certifies LTT and the best of T1/T2 under three alternative harm definitions: collision only (no time-to-collision term, base rate 3.2 %, $\alpha = 0.01$) and time-to-collision thresholds of 0.5 s and 1.5 s instead of the headline 0.95 s. The geometric score’s advantage survives every definition: the paired difference in unnecessary stops excludes 0 by at least 20 points in each case.

Does validity replicate on unseen data and a second predictor? We recalibrate LTT on 904 fresh Argoverse 2 scenarios, disjoint from every split above, and test on a further 904, then repeat this with a second, independently trained predictor (minADE₆ = 0.854, versus 1.09 for the predictor used everywhere else). T3 and ACI are not available on this data (no open-loop error files) and are omitted rather than approximated, and T2 is added by re-running the reference

TABLE V

LTT (GEOMETRIC) VERSUS THE BETTER OF T1/T2 UNDER ALTERNATIVE HARM DEFINITIONS, ARGOVERSE 2 TEST. UNNECESSARY-STOP RATES IN %; “BEST” IS THE BETTER OF T1/T2 AMONG THOSE MEETING THE MISS TARGET, AND THE PAIRED 95 % CI IS OF THE STOP-RATE DIFFERENCE.

Harm definition	LTT	Best	Which	Diff. [95 % CI]
Collision only, $\alpha=0.01$	45.1	83.9	T2	-38.8 [-41.4, -36.2]
TTC < 0.5 s	15.8	38.6	T1	-22.7 [-25.5, -20.1]
TTC < 0.95 s (headline)	28.5	68.3	T1	-39.8 [-42.4, -37.2]
TTC < 1.5 s	34.0	68.3	T2	-34.4 [-36.8, -31.9]

runs with dropout. LTT’s validity replicates in both settings (7.5 % and 9.0 % violations, against 46–51 % for the uncertified baselines), and its miss rate stays below α (Supplementary Section S8).

VI. SCOPE AND FUTURE WORK

What we claim, and what we do not. We claim that certification is what makes a closed-loop trigger keep its alert-level miss rate, at a measurable cost in stops, and that the guarantee is fragile to shift and latency. We do not claim a better-performing trigger (the pre-registered hypothesis that said so is refuted) or a guarantee on the outcome.

Alert versus outcome. The certificate bounds the probability, over scenarios, that harm is not pre-empted by an alert. A deployment authority can audit it, but it is weaker than a bound among harmful scenarios (24 % go unalerted, 58 % on nuScenes) and it bounds the alert, not the outcome. Certifying residual harm directly was not achievable with this score and maneuver (Supplementary Section S5), so it calls for a different maneuver or score rather than a different calibrator. The 5.0 % first-tick floor on nuScenes also shows that a miss metric on harm anywhere in a replayed snippet mixes preventable events with events already under way, and we propose an avoidable-only variant for future benchmarks.

Simulator, predictor and perception. All results use MetaDrive with rule-based (IDM) agents, so whether the ordering of triggers survives learned agent behavior is open, and the induced-collision rate, which depends on how that traffic reacts to a stationary ego, is best read as an upper bound. LTT’s validity and the gap to the confidence and ensemble triggers replicate with a second AutoBot predictor (Section V-G), and validity and the confidence gap also hold with a different architecture, Wayformer (7.0 % violations, 41.8 % against 67.8 % stops), at a higher stop rate (Supplementary Section S13). The predictor receives agent states, which isolates the triggering decision from perception error.

Shift and dependence. The weighted certificate reports when it becomes unreliable through its effective sample size (68 in our nuScenes subset). All guarantees treat scenarios as exchangeable, which nearby scenarios may violate: our nuScenes samples span only 37 and 23 scenes, and spatial-group resplits (a proxy) push the violation frequency above δ , so the high-probability statement is weaker in practice.

Harm definition and fallback family. The geometric score’s advantage holds under three time-to-collision thresholds and a collision-only definition (Section V-G). Fault attribution is not varied, and a severity-weighted or tail certificate [28]

requires only a change of loss. We evaluate in-lane stops at the UN R157 deceleration limit, and other fallbacks change the quantities priced but not the method.

VII. CONCLUSION

An alert-level guarantee on a fallback trigger is not a permanent property of the model. In the setting studied, it depends on the data it was certified on, the timing of the intervention, and what the fallback itself does. Evaluated with the fallback executed, a Learn-then-Test certified trigger keeps its 5 % target on the calibration distribution (3.7 % missed, violations in 9.5 % of resplits against 43.5 % without a certificate) at a cost of about 2.5 points of unnecessary stops, and deployed-practice confidence and ensemble triggers need 68–71 % stops for similar safety. The certificate controls the alert, not the outcome: harm in the intervened run is 6.5 %, 1.1 % of it created by the intervention, and a 5 % marginal target leaves about a quarter of harmful scenarios unalerted. The same certificate fails on nuScenes (13.2 % missed, with wide uncertainty) and in two of six held-out Argoverse 2 cities, both with high harm rates, but holds, conservatively, on Waymo (2.7 % missed), so failure under shift is dataset-dependent. One decision tick of latency breaks it, and it can be unreachable when the target data contain harm no trigger can prevent. These findings argue, for the setting studied, for treating certification, shift monitoring and latency as one audited quantity, and the lookup protocol makes that audit inexpensive to repeat for any trigger.

ACKNOWLEDGMENT

The authors thank the maintainers of MetaDrive and ScenarioNet for releasing the simulation infrastructure this work builds on.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Berlin, Germany: Springer-Verlag, 2005.
- [2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, “Learn then test: Calibrating predictive algorithms to achieve risk control,” *Ann. Appl. Statist.*, vol. 19, no. 2, pp. 1641–1662, 2025.
- [3] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
- [4] L. Lindemann, M. Cleaveland, G. Shim, and G. J. Pappas, “Safe planning in dynamic environments using conformal prediction,” *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, Aug. 2023.
- [5] A. Dixit, L. Lindemann, S. X. Wei, M. Cleaveland, G. J. Pappas, and J. W. Burdick, “Adaptive conformal prediction for motion planning among dynamic agents,” in *Proc. 5th Annu. Conf. Learn. Dyn. Control (LADC)*, PMLR vol. 211, 2023, pp. 300–314.
- [6] B. Wilson *et al.*, “Argoverse 2: Next generation datasets for self-driving perception and forecasting,” in *Proc. 35th Conf. Neural Inf. Process. Syst. (NeurIPS)*, Track Datasets and Benchmarks, 2021.
- [7] H. Caesar *et al.*, “nuScenes: A multimodal dataset for autonomous driving,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11621–11631.

- [8] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E*, vol. 62, no. 2, pp. 1805–1824, 2000.
- [9] A. N. Angelopoulos, R. Foygel Barber, and S. Bates, "Theoretical foundations of conformal prediction," Cambridge Univ. Press (to appear), *arXiv preprint arXiv:2411.11824*, 2024.
- [10] L. Lindemann, Y. Zhao, X. Yu, G. J. Pappas, and J. V. Deshmukh, "Formal verification and control with conformal prediction," *arXiv preprint arXiv:2409.00536*, 2024.
- [11] Y. Shu, J. Xu, L. Kong, F. Zhang, G. Yin, and C. Sun, "Scenario-aware uncertainty quantification for trajectory prediction with statistical guarantees," *arXiv preprint arXiv:2512.05682*, 2025.
- [12] H. Wang and C. Ning, "Conformal prediction in the loop: A feedback-based uncertainty quantification model for trajectory optimization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.
- [13] P. Y. Chang and Q. Ahmed, "Barrier function conformal safety clearance certification with CVaR for driving trajectory selection," *arXiv preprint arXiv:2608.26533*, 2026.
- [14] C. Guo *et al.*, "CoCoNav: Conformal control for safe robot navigation in crowds," *arXiv preprint arXiv:2608.07751*, 2026.
- [15] A. N. Angelopoulos, "Conformal risk control for non-monotonic losses," *Phil. Trans. R. Soc. A*, 2026, doi: 10.1098/rsta.2025.0068.
- [16] S. Joshi, Y. Sun, H. Hassani, and E. Dobriban, "MultiRisk: Multiple risk control via iterative score thresholding," *arXiv preprint arXiv:2512.24587*, 2025.
- [17] J. Gonzales, K. Mizuta, K. Leung, and L. J. Ratliff, "Safe probabilistic planning for human-robot interaction using conformal risk control," *arXiv preprint arXiv:2603.10392*, 2026.
- [18] S. Joshi, T. Wang, H. Hassani, and E. Dobriban, "Risk-controlled post-processing of decision policies," *arXiv preprint arXiv:2605.06479*, 2026.
- [19] H. Atoui and P. Resende, "Minimum risk maneuver fallback strategy for autonomous vehicles: Design and experimental validation," in *Proc. IEEE Intell. Veh. Symp. (IV)*, 2025, pp. 1979–1984, doi: 10.1109/IV64158.2025.11097617.
- [20] X. Zhao, R. Aghazadeh-Chakherlou, C.-H. Cheng, P. Popov, and L. Stigini, "On the need for a statistical foundation in scenario-based testing of autonomous vehicles," in *Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC)*, 2025.
- [21] T. Puphal, V. Ramtekkar, and K. Nishimiya, "Risk-based filtering of valuable driving situations in the Waymo Open Motion Dataset," *arXiv preprint arXiv:2506.23433*, 2025.
- [22] J. Shi *et al.*, "Motion forecasting for autonomous vehicles: A survey," *arXiv preprint arXiv:2502.08664*, 2025.
- [23] K. Rahaman, J. V. Deshmukh, A. R. Hota, and L. Lindemann, "When environments shift: Safe planning with generative priors and robust conformal prediction," in *Proc. 8th Annu. Conf. Learn. Dyn. Control (LADC)*, 2026.
- [24] A. Doula, M. Mühlhäuser, and A. S. Guinea, "SafePath: Conformal prediction for safe LLM-based autonomous navigation," *arXiv preprint arXiv:2505.09427*, 2025.
- [25] A. Farid, S. Veer, B. Ivanovic, K. Leung, and M. Pavone, "Task-relevant failure detection for trajectory predictors in autonomous vehicles," in *Proc. 6th Conf. Robot. Learn. (CoRL)*, Auckland, New Zealand, 2022.
- [26] S. Ettinger *et al.*, "Large scale interactive motion forecasting for autonomous driving: The Waymo Open Motion Dataset," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, arXiv:2104.10133.
- [27] R. Luo, S. Zhao, J. Kuck, B. Ivanovic, S. Savarese, E. Schmerling, and M. Pavone, "Sample-efficient safety assurances using conformal prediction," *Int. J. Robot. Res.*, vol. 43, no. 9, pp. 1409–1424, 2024.
- [28] M. P. Chapman, R. Bonalli, K. M. Smith, I. Yang, M. Pavone, and C. J. Tomlin, "Risk-sensitive safety analysis using conditional value-at-risk," *IEEE Trans. Autom. Control*, vol. 67, no. 12, pp. 6521–6536, Dec. 2022.
- [29] Q. Li, Z. Peng, L. Feng, Z. Liu, C. Duan, W. Mo, and B. Zhou, "ScenarioNet: Open-source platform for large-scale traffic scenario simulation and modeling," in *Proc. 37th Conf. Neural Inf. Process. Syst. (NeurIPS)*, Track Datasets and Benchmarks, 2023.
- [30] N. Karnchanachari *et al.*, "Towards learning-based planning: The nuPlan benchmark for real-world autonomous driving," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [31] J. Lekeufack, A. N. Angelopoulos, A. Bajcsy, M. I. Jordan, and J. Malik, "Conformal decision theory: Safe autonomous decisions from imperfect predictions," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 11668–11675.
- [32] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.
- [33] R. J. Tibshirani, R. Foygel Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019.
- [34] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, 2016, pp. 1050–1059.
- [35] R. Girgis, F. Golemo, F. Codevilla, M. Weiss, J. A. D'Souza, S. E. Kahou, F. Heide, and C. Pal, "Latent variable sequential set transformers for joint multi-agent motion prediction," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [36] L. Feng, M. Bahari, K. M. Ben Amor, É. Zablocki, M. Cord, and A. Alahi, "UniTraj: A unified framework for scalable vehicle trajectory prediction," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024.
- [37] United Nations Economic Commission for Europe, "UN Regulation No. 157 — Uniform provisions concerning the approval of vehicles with regard to Automated Lane Keeping Systems," Rev. 1, 2025.
- [38] R. Sinha, E. Schmerling, and M. Pavone, "Closing the loop on runtime monitors with fallback-safe MPC," in *Proc. IEEE Conf. Decis. Control (CDC)*, 2023.
- [39] S. Sohrabi *et al.*, "Quantifying the automated vehicle safety performance: A scoping review of the literature, evaluation of methods, and directions for future research," *Accid. Anal. Prev.*, vol. 152, Art. no. 106003, 2021.
- [40] D. Xie *et al.*, "A survey on scenario-based test and evaluation methods for autonomous vehicles," *Proc. Inst. Mech. Eng. D, J. Automobile Eng.*, 2025.
- [41] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "On a formal model of safe and scalable self-driving cars," *arXiv preprint arXiv:1708.06374*, 2017, doi: 10.48550/arXiv.1708.06374.
- [42] International Organization for Standardization, "ISO 21448:2022, Road vehicles—Safety of the intended functionality (SOTIF)," ISO, 2022.
- [43] UL Standards & Engagement, "UL 4600, Standard for safety for the evaluation of autonomous products," 3rd ed., Mar. 2023.
- [44] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017, pp. 4878–4887.



Hassan Waqas Saleem received the M.S. degree in electrical engineering, with a specialization in artificial intelligence and autonomous systems, from the National University of Sciences and Technology (NUST), Pakistan, in 2024, the B.Sc. degree in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2018, and the B.Tech. degree in electrical technology from Bahauddin Zakariya University, Multan, Pakistan, in 2014. He is currently a Technical Lead with a public-sector organization in Pakistan. His research interests include artificial intelligence for autonomous systems, uncertainty quantification and safety assurance for automated driving, graph neural networks, knowledge graph completion, large language models, robotics, and simultaneous localization and mapping.

Rahat Zeeshan Saleem received the M.B.A. degree from Bahauddin Zakariya University, Multan, Pakistan, in 2021, where he was awarded a gold medal, and the M.S. degree in supply chain management from the NUST Business School, National University of Sciences and Technology (NUST), Islamabad, Pakistan, in 2023. He is currently a Project Lead with a private-sector organization in Pakistan. His research interests include artificial intelligence and data-driven decision support.

Hina Maryam received the M.Phil. degree in statistics from Bahauddin Zakariya University, Multan, Pakistan, in 2019, where she was awarded a gold medal. She is currently an Assistant Director with the State Bank of Pakistan, Karachi, Pakistan. Her research interests include statistical modeling, data optimization, and applied data analysis.